Y. Jayesh Ranjan
192472201

UNIT 3 – HANDS-ON EXERCISES

Step-by-Step Solutions

1. Mean and Population Variance

Given:

$$X = 20, 25, 30, 35, 40$$

Mean

$$\boxed{\bar{X} = \Sigma X / N}$$

$$\Sigma X = 20 + 25 + 30 + 35 + 40$$

$$= 150$$

$$N = 5$$

$$\bar{X} = 150 / 5$$

$$\boxed{\textbf{Mean = 30 minutes}}$$

Population Variance

X	**X − 30**	**(X − 30)²**
20	-10	100
25	-5	25
30	0	0
35	5	25
40	10	100
Total		250

$$\boxed{\sigma^2 = \Sigma(X - \bar{X})^2 / N}$$

$$\sigma^2 = 250 / 5$$

$$\boxed{\textbf{Population Variance} = \textbf{50 minutes}^2}$$

Answer:

Mean = 30 minutes

Population Variance = 50 minutes^2

2. Sampling

Given:

Population, N = 5000 students

Sample, n = 100 students

Percentage of students surveyed:

$$(100 / 5000) \times 100$$

$$= 10000 / 5000$$

$$\boxed{= \mathbf{2\%}}$$

Answer:

Sampling is used because it saves time, cost and effort while providing an estimate of the population average.

3. Sample Proportion

Given:

Total customers = 200

Satisfied customers = 150

$$\boxed{\hat{p} = x / n}$$

$$\hat{p} = 150 / 200$$

$= 3 / 4$

Sample Proportion = 0.75

Percentage:

$$0.75 \times 100 = 75\%$$

Percentage of satisfied customers = 75%

4. Covariance Between X and Y

X	**Y**
2	5
4	7
6	9
8	11

Mean of X:

$$\bar{X} = (2 + 4 + 6 + 8) / 4 = 20 / 4 = 5$$

$\boldsymbol{\bar{X} = 5}$

Mean of Y:

$$\bar{Y} = (5 + 7 + 9 + 11) / 4 = 32 / 4 = 8$$

$\boldsymbol{\bar{Y} = 8}$

X	**Y**	**X − 5**	**Y − 8**	**(X − 5)(Y − 8)**
2	5	-3	-3	9
4	7	-1	-1	1
6	9	1	1	1
8	11	3	3	9
				20

$$\Sigma(X - \bar{X})(Y - \bar{Y}) = 20$$

$Cov(X,Y) = \Sigma(X - \bar{X})(Y - \bar{Y}) / N$

$= 20 / 4 = 5$

Covariance = 5

Relationship = Positive

5. Covariance – Exercise Hours and Calories

X (Exercise Hours)	Y (Calories)
1	200
2	300
3	400
4	500
5	600

Mean of X:

$$\bar{X} = (1 + 2 + 3 + 4 + 5) / 5 = 15 / 5 = 3$$

$\bar{X} = 3$

Mean of Y:

$$\bar{Y} = (200 + 300 + 400 + 500 + 600) / 5 = 2000 / 5 = 400$$

$\bar{Y} = 400$

X	Y	X − 3	Y − 400	Product
1	200	-2	-200	400
2	300	-1	-100	100
3	400	0	0	0
4	500	1	100	100
5	600	2	200	400
				1000

$$\Sigma(X - \bar{X})(Y - \bar{Y}) = 400 + 100 + 0 + 100 + 400 = 1000$$

$$\boxed{\mathbf{Cov(X,Y) = 1000 / 5}}$$

$$\boxed{\mathbf{Covariance = 200}}$$

$$\boxed{\mathbf{Relationship = Positive}}$$

6. Pearson's Correlation Coefficient

X (Study Hours)	**Y (Marks)**
1	45
2	50
3	55
4	65
5	70

X	**Y**	**X²**	**Y²**	**XY**
1	45	1	2025	45
2	50	4	2500	100
3	55	9	3025	165
4	65	16	4225	260
5	70	25	4900	350
15	285	55	16675	920

Therefore:

$$\Sigma X = 15, \quad \Sigma Y = 285, \quad \Sigma X^2 = 55, \quad \Sigma Y^2 = 16675, \quad \Sigma XY = 920, \quad N = 5$$

$$\boxed{\mathbf{r = [N\Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{\{[N\Sigma X^2 - (\Sigma X)^2][N\Sigma Y^2 - (\Sigma Y)^2]\}}}}$$

Numerator:

$$5(920) - (15)(285)$$

$$= 4600 - 4275$$

$$\boxed{\mathbf{= 325}}$$

First denominator term:

$$5(55) - (15)^2 = 275 - 225 = 50$$

$$\boxed{\mathbf{= 50}}$$

Second denominator term:

$$5(16675) - (285)^2 = 83375 - 81225 = 2150$$

$$\boxed{\mathbf{= 2150}}$$

Complete denominator:

$$\sqrt{(50 \times 2150)} = \sqrt{107500} \approx 327.8719$$

$$\boxed{\textbf{Denominator} \approx \mathbf{327.8719}}$$

Final calculation:

$$r = 325 / 327.8719$$

$$\boxed{\mathbf{r \approx 0.9912 \approx 0.991}}$$

$$\boxed{\textbf{Strength = Very Strong}}$$

$$\boxed{\textbf{Direction = Positive}}$$

Interpretation:

There is a very strong positive correlation between study hours and marks. As study hours increase, marks also tend to increase.

FINAL ANSWERS

Q.No.	**Result**	**Final Answer**
1	Mean	30 minutes
1	Population Variance	50 minutes^2
2	Sampling	Saves time, cost and effort
3	Sample Proportion	0.75
3	Percentage	75%
4	Covariance	5
4	Relationship	Positive
5	Covariance	200
5	Relationship	Positive
6	Pearson Correlation	$r \approx 0.991$
6	Strength	Very strong positive correlation